Response to Reviewers

VOUCH: Anytime-Valid Certificates for Machine Unlearning via Paired-Canary Hypothesis-Betting

Machine Learning — revised submission

We thank both reviewers for reports that were unusually careful and, in two places, mathematically decisive. Reviewer 1's observation that Definition 1's marginal null does not match the condition our supermartingale proof actually uses is correct, and acting on it has changed the paper's central theorem rather than merely its wording. Reviewer 1's correction of the Kullback–Leibler expansion is likewise correct. Reviewer 2's insistence that every end-to-end result sat at a permissive tolerance, that the declared score class was never stress-tested from outside, that the $\varepsilon = 0.02$ entries in Table 3 were censoring artefacts, and that we had asserted the canaries "do not stand out" without testing it, were each right, and in the last case the measurement we now report contradicts what we had written. We have made every change requested. Where a request was not fully satisfiable within the compute available we say so explicitly rather than substituting a weaker experiment.

A note on numbering. The revision adds five experiment subsections, six tables, one appendix and three theory statements, so section, theorem, remark, table and figure numbers differ between the two versions. Numbers inside the quoted reviewer points refer to the previous version; numbers in our responses refer to the revised manuscript unless we say otherwise. The correspondences a reader of both versions will meet most often are: Theorem 2 (certificate coverage) is still Theorem 2, restated; Theorem 3 (power) is now Theorem 4; the finite-cohort power result added at this revision is Theorem 5, which shifts Theorem 4 (streaming) to Theorem 6 and Proposition 6 (bridge) to Proposition 8; Table 3 (power against the Kullback–Leibler limit) is now Table 6; Table 5 (GPT-2 end-to-end) is now Table 9; Table 7 (model axis) is now Table 18; and the experiments remain Section 5.

Summary of the substantive changes.

1. **The certificate's null is restated and its theorem replaced.** We now certify the *realised advantage on the committed cohort* and bet against a without-replacement boundary recentred at every step. The resulting guarantee (Theorem 2) is exact under arbitrary dependence across pairs, templates and repetition strata — no independence, no exchangeability across pairs, no homogeneity. A new Proposition 3 prices the transfer to the super-population advantage under a named assumption. Both targets are reported side by side throughout.

2. **New simulation showing the old null was genuinely not controlled** (§5.1, Tables 1–2): under model-level over-dispersion the previous fixed-boundary process issues under the marginal null in 38.9% of runs, an eightfold breach of α; the corrected process holds at ≤ 0.020 in every regime. Heterogeneity is also measured on the real runs.

3. **Four valid sequential comparators added** (§5.4, Table 5): exact group-sequential α-spending (Pocock, O'Brien–Fleming), a Jeffreys-mixture e-process, a Robbins normal-mixture e-process, and a fixed-n test at a pre-committed size.

4. **Tolerance sweep to $\varepsilon = 0.05$ on real models** (Table 11), with the cohort/super-population distinction made explicit; guidance on choosing ε added.

5. **Mathematical corrections**: $\mathrm{KL} = \varepsilon^2/2 + O(\varepsilon^4)$, so the cohort rule is $2\log(1/\alpha)/\varepsilon^2$; the score sign convention fixed to the token-normalised *log-likelihood*; per-score versus simultaneous confidence-sequence coverage stated; Proposition 8's one-directionality stated immediately after it.

6. **Statistical reporting**: Clopper–Pearson intervals on every verdict count, across-seed dispersion on every mean column, censoring rates and Kaplan–Meier medians in the power table, seed counts on the model axis, and the "stable across the axis" characterisation corrected.

7. **Three new empirical sections**: head-to-head against forget-quality and min-$k\%$ on identical runs (§5.12); attacks from *outside* the declared class (§5.13); canary detectability, with the negative result and the entropy-dilution design rule that follows from it (§5.16).
8. **Four further additions made after the first revision draft**, each answering a request we had previously answered only partially: a LiRA-style shadow-model attack with matched shadows and a positive control (§5.14); RMU as a second utility-preserving subject, chosen because it never touches the token loss (§5.9); a paraphrase-aware score added to $\mathcal{F}$ (§5.15); and Theorem 5, a finite-cohort certification-time bound that supplies the finite-population power analysis Theorem 4 could not.
9. **Verifier-side reveal randomness implemented**: the released code now draws the reveal order from a public randomness beacon, binding it to the manifest commitment and recording the round on the certificate, with operating-system entropy as the offline fallback (Remark 1, Appendix A).
10. **SimNPO added** as a subject; **benchmark forget/retain and general-capability readings** added; **reproducibility appendix** added, including an unambiguous definition of "retraining".
11. **Scope made consistent** in abstract, introduction, experiments and conclusion; legal claims sourced to primary instruments; the 99.6% figure attributed to our own earlier design; promotional phrasing removed; the six papers requested are discussed, not merely cited.

Reviewer 1

1. The null in Definition 1 does not match the condition used in Theorem 2

R1.1

Definition 1 defines $p^{(s)} = \Pr[Z_i^{(s)} = 1]$, a marginal probability, whereas the proof assumes $\mathbb{E}[Z_i^{(s)} \mid \mathcal{F}_{i-1}] \geq p_0$ at every step, which is stronger. The signs may not be independent because all canaries pass through the same jointly trained and unlearned model, and may differ across templates and repetition strata. Please either redefine the certificate using an appropriate conditional-mean null and justify why the setting satisfies it, or provide a result valid under the dependence structure considered here. It would also help to state which sources of randomness define $p^{(s)}$. Simulations with dependent or heterogeneous signs would strengthen this part.

Response. This is correct, it is the most consequential point in either report, and we have taken the second of the two options offered — a result valid under the dependence structure — because we agree the first would have been a patch. Four changes follow.

(a) Which randomness defines the null (§3.3, §4). The randomness is the *inclusion coins*, and we now make the filtration carry that literally. The manifest is committed as a Merkle tree with one leaf per pair, and leaf $\pi(t)$ is opened at step t, so the coin $b_{\pi(t)}$ is genuinely unrevealed at time $t-1$. Theorem 1 is restated against this filtration and now reads: under exact unlearning, $Z_{\pi(t)}^{(s)} \mid \mathcal{F}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$ *exactly*, **whether or not the signs are independent across pairs**. The proof never asks for independence; it asks only that the coin governing the pair opened now still be fair given everything opened before, which under exact unlearning it is. The revocation arm therefore needs no repair — the reviewer's concern does not bite there, and we now say so explicitly.

(b) The certificate is retargeted (Definition 2, Theorem 2). The composite null "some score still leaks" is where dependence does bite. Once M_u and the manifest exist, the cohort's sign vector is *fixed*; the only randomness left to the audit is the reveal order. We therefore certify

$$\Delta^{(s)} = \frac{2}{m} \sum_{i=1}^{m} z_i^{(s)} - 1,$$

the *realised cohort advantage* — what the declared score actually achieves against the committed cohort, which is what an auditor of *this* model wants bounded. Sequential reveal of a committed uniform permutation is simple random sampling without replacement, so we bet against the re-centred boundary $p_0(t) = (mp_0 - \sum_{u<t} Z_{\pi(u)})/(m-t+1)$, the mean the unrevealed remainder would need for the null to hold. Theorem 2 shows this is a nonnegative supermartingale *conditionally on the sign vector*, so its coverage holds under **arbitrary** dependence across pairs, arbitrary heterogeneity across templates and repetition strata, and arbitrary non-identical distribution. No assumption whatsoever is spent on how the sign vector arose.

(c) The super-population claim is available but priced (Proposition 3). Under a named *bounded coin influence* assumption — flipping one coin changes at most κ signs, with $\kappa = 1$ automatic under exact unlearning — McDiarmid gives $|\Delta^{(s)} - \bar{\Delta}^{(s)}| \leq \kappa\sqrt{2\log(2/\alpha')/m}$. At $m = 512$, $\alpha' = 0.05$, $\kappa = 1$ that is 0.12. So the algorithm-level statement costs a tolerance inflation we now state, and this is exactly why our tight-tolerance end-to-end results are cohort certificates (see R2.W1).

(d) Simulations with dependent and heterogeneous signs (§5.1, Tables 1–2). We built the study you asked for and it vindicates the concern. Sign streams with *marginal* mean exactly $p_0 = 0.55$ under four regimes, 2,000 seeds, peeking after every pair:

regime	sd($\hat{\Delta}$)	fixed boundary (v1)	without replacement (ours)
independent	0.031	0.034	0.014
heterogeneous by stratum	0.030	0.026	0.020
model-level over-dispersion	0.296	**0.389**	0.001
shared model-level latent	0.500	**0.481**	0.000

Independence and stratum heterogeneity alone leave the old process valid. A shared model-level random effect — one training run leaks, another does not — breaks it eight- to tenfold. The corrected process controls the realised-cohort error at ≤ 0.020 everywhere. Table 2 verifies Theorem 2 directly on *fixed* adversarially structured sign vectors at the least favourable feasible point of the cohort null: error never exceeds 0.035 against nominal 0.05, for every structure, both tolerances, both cohort sizes.

Finally, the dependence is real on our own runs: a chi-square test of sign-rate homogeneity across repetition strata rejects in 24 of the 32 un-unlearned runs, where there is leakage to be heterogeneous about, and in 14% of runs overall.

We note in §4.2 one honest consequence of the retargeting: a cohort audited to exhaustion resolves deterministically, so the betting machinery buys early stopping, validity under continuous monitoring, and the alarm arm, rather than buying the cohort statement itself. We prefer to state that than to leave the reader to discover it.

2. Scope of the certificate

R1.2

VOUCH certifies the advantage of a declared score class on the planted canary population, which is not the same as proving all organic forget records were removed. Some statements, such as "certifies genuine unlearning," sound stronger than the guarantee. Make the scope consistent throughout. Provide more direct evidence that the canary results track forgetting on organic records. Relatedly, the commitment does not prevent a provider who knows the canaries from treating them differently; state the honest-but-verifiable assumption whenever the certification claim is summarised.

Response. Agreed on all three counts, and acted on.

Consistency. "Certifies genuine unlearning" is gone. The abstract now closes with "The certificate concerns a declared score class on a committed cohort under an honest-but-verifiable provider, and we measure what each qualification costs." The introduction has a dedicated paragraph, *What the certificate claims, and what it does not*, listing the four qualifications with the measured cost of each. The theory section now closes with a paragraph titled *Scope, stated precisely* (end of §4). The conclusion enumerates the same four with their costs. We also checked every remaining occurrence of "certif*" in the manuscript for an unqualified claim.

Evidence that canaries track organic forgetting. We have added the comparison you suggest, in the strongest form our data supports (§5.12, Table 17): on identical models and seeds we compute a TOFU-style forget-quality analogue (two-sample KS between the unlearned and retrained models' canary-score distributions) and a MUSE-style min-k% leakage analogue (paired AUC relative to the retrained reference), and report them beside VOUCH's verdicts. Three disagreements emerge, and we report them in both directions. The forget-quality analogue is systematically *pessimistic*, failing genuine unlearning methods whose realised advantage is within noise of zero, because a KS test detects distributional differences far below any tolerance one would certify — it answers "are these identical" when the question is "is the residual advantage below ε". PrivLeak reads -2.4% on the MUSE/Phi-1.5 GradDiff seed that VOUCH *revokes* (-1.2% averaged over that cell's 3 seeds), which is the concrete instance of a claim the previous version made only rhetorically. And min-k% inspected after every pair flags leakage on genuinely clean retrained models in several cells. We are also even-handed about the limits: these are analogues computed on canary cohorts, not the benchmarks' own metrics on their own splits, and we say so.

We have additionally strengthened the dose–response reporting, since it is the only calibration of the canary-to-organic extrapolation we can honestly offer. Mean score gaps rise monotonically with training-time repetition (+0.46, +0.78, +1.54, +2.35 nats at $r = 1, 2, 4, 8$, over the six benchmark/model cells) and so do realised advantages (+0.22, +0.35, +0.59, +0.73). We now also state the limit this implies: the $r = 1$ stratum, the one most like an organic record, carries the *weakest* signal, so the extrapolation is weakest exactly where it matters most. We do not claim a transfer theorem, because we do not have one.

The threat model. The honest-but-verifiable assumption is now stated in §3.1, in the introduction's scope paragraph, in the scope paragraph that closes §4, and in the conclusion — i.e. wherever the certification claim is summarised. It also acquired teeth we had not expected: see our response to R2.8, where a measurement shows a zero-knowledge perplexity filter recovers the *entire* planted cohort. We retract the manuscript's previous claim that the canaries "do not stand out" and now explain both why detection does not by itself defeat the audit (both twins are equally conspicuous, so locating the cohort reveals nothing about which twin the coin admitted) and why closing the remaining gap requires cryptographic attestation, which we do not supply.

3. Mathematical and notational inconsistencies

R1.3a — sign convention

Eq. (2) defines $D_i = s(M_u, c_{\text{in}}) - s(M_u, c_{\text{ghost}})$, but Fig. 5 appears to use the reverse sign: NLL values 0.99/9.03 are reported as $D = +8.0$, which under Eq. (2) should have the opposite sign if s is NLL. Use one convention throughout theory, figures, tables and implementation.

Response. You identified a real inconsistency, and we can now say precisely where it was. The released implementation's score is `s_loss = mean(logprobs)`, i.e. the token-normalised *log-likelihood* — the *negative* of the NLL. Under that definition $D = (-0.99) - (-9.03) = +8.04$, so Eq. (2), Fig. 5, the tables and the code were all mutually consistent; what was wrong was the *prose* in the previous version's §4.2, which described the score as "the token-normalised negative log-likelihood". That single word inverted the stated convention.

We have fixed the prose rather than the figure, and pinned the convention down so it cannot drift again. §3.4 now lists every member of $\mathcal{F}$ with the orientation stated explicitly ("each oriented so that a *higher* score means *more memorised*"), defines s_{loss} as the token-normalised log-likelihood, and states the convention at the point where it could otherwise be misread. Equation (2) is now followed by the reading rule: a positive D means the trained twin is the better-remembered one, which is the direction of leakage. Figure 5's caption and body now restate the convention at the point of use. We verified the +8.0 figure against the released pipeline output.

R1.3b — the small-ε expansion

$\mathrm{KL}(\mathrm{Bern}(1/2)\|\mathrm{Bern}(1/2+\varepsilon/2)) = -\frac{1}{2}\log(1-\varepsilon^2) = \varepsilon^2/2 + O(\varepsilon^4)$*, so the leading approximation for certification time should be about $2\log(1/\alpha)/\varepsilon^2$, not $8\log(1/\alpha)/\varepsilon^2$. Interestingly the corrected expression is consistent with Table 3 and with "about 600 pairs for $\varepsilon = 0.1$, $\alpha = 0.05$". Please correct the theorem and recheck related calculations.*

Response. You are exactly right, including in your inference about Table 3 (now Table 6). Theorem 4 now carries the derivation in full:

$$\mathrm{KL}\big(\mathrm{Bern}(\tfrac{1}{2}) \,\|\, \mathrm{Bern}(p_0)\big) = -\tfrac{1}{2}\log\big(4p_0(1-p_0)\big) = -\tfrac{1}{2}\log(1-\varepsilon^2) = \tfrac{\varepsilon^2}{2} + O(\varepsilon^4),$$

so $\mathbb{E}[\tau^*] \approx 2\log(1/\alpha)/\varepsilon^2$. We rechecked every dependent number. The tables were computed in code from the *exact* divergence (`kl = 0.5*log(0.5/p0) + 0.5*log(0.5/(1-p0))`), so no numerical column changes: at $\varepsilon = 0.1$, $\alpha = 0.05$ the exact bound is 596 and $2\log(1/\alpha)/\varepsilon^2$ gives 599, against the erroneous rule's 2,397. Only the stated rule of thumb was wrong, by a factor of four. Theorem 4 now says so in a sentence, so a reader of the previous version is not left guessing. The cohort-sizing prose is updated (600 pairs at $\varepsilon = 0.1$, 2,400 at 0.05, 15,000 at 0.02), as is the cost discussion, which previously quoted the wrong scaling.

R1.3c — per-score or simultaneous confidence sequences?

Clarify whether the confidence-sequence guarantee is per score or simultaneous over the whole score class $\mathcal{F}$.

Response. Added as Remark 3, distinguishing three statements. Each score's sequence covers *its own* advantage with uniform-in-time probability $1-\alpha$: the guarantee is per score. The reported $\max_{s\in\mathcal{F}} U_t^{(s)}$ *is* nonetheless a valid level-α anytime upper bound on $\sup_{s\in\mathcal{F}} \Delta^{(s)}$ with no multiplicity correction, because once the cohort is fixed the maximising index of the population quantities is a fixed, non-random index, and only that one score's sequence is needed. Simultaneous two-sided coverage of *all* $|\mathcal{F}|$ advantages does cost a union bound and holds at $|\mathcal{F}|\alpha$; we now report intervals at that corrected level wherever the whole class is displayed.

4. A fairer optional-stopping comparison

R1.4

The fixed-sample binomial test inspected after every pair is a useful illustration but not a strong baseline. Please add at least one valid sequential baseline — alpha-spending, group-sequential, or another Bernoulli confidence-sequence/e-process method — under the same observation stream and query budget. A standard fixed-n comparison at a precommitted sample size would also help show the practical cost of anytime validity.

Response. Agreed; the peeking binomial was a demonstration, not a competitor, and we have added all four comparators you name (§5.4, Table 5), on the same sign stream and the same query budget:

- exact group-sequential α-spending with Pocock and O'Brien–Fleming spending functions, with boundaries calibrated by *dynamic programming over the exact binomial lattice* at the least favourable null rather than by a normal approximation, so the baseline is as strong as it can be made;
- a truncated Beta($\frac{1}{2}, \frac{1}{2}$)-mixture (Jeffreys) e-process;
- a Robbins normal-mixture e-process;
- a fixed-n binomial test at a pre-committed sample size.

One presentational error we made in a first pass and have fixed, because it mattered: the table's type-I column originally printed VOUCH's cohort arm on a *conditional* error rate — restricted to runs whose realised cohort advantage violates the tolerance — and the six comparators on the *marginal* rate. That is not a comparison, and it flattered us by a factor of five. Both statistics are now printed for all seven rows. Against the marginal null the super-population procedures all hold level (0.030–0.053 at 2,000 seeds, standard error ≈ 0.005) and VOUCH's cohort arm does not, by construction, since half of those cohorts have a realised advantage genuinely below the tolerance; against the cohort null all seven hold level, and the group-sequential and fixed-n procedures are markedly conservative there because they were built for the other target. Neither column alone is a fair scoreboard, so we print both and the text says which answers which question.

On power and cost, against exact unlearning at $\varepsilon = 0.1$ with a budget of 1,536 pairs: VOUCH (cohort) certifies 99.95% of runs using a mean of 523 queries; Pocock 97.1% using 645; O'Brien–Fleming 98.5% using 844; the mixture e-processes 77–78%; the fixed-n test 98.9% but spending all 1,536 queries on every run, since it cannot stop early.

The honest summary, which we now give in the text, is that anytime validity is close to free here: against the strongest valid sequential comparator VOUCH gives up nothing in power and saves about a fifth of the queries, and it matches fixed-n's power to within a rounding of one point at a threefold reduction in verifier cost. The table also isolates how much of the advantage comes from our retargeting rather than from betting: on the super-population target, same budget, power falls from 1.000 to 0.882 and mean queries rise from 523 to 747 — the same price Proposition 3 charges in tolerance. We think separating those two effects is more useful than a single headline number.

5. Stronger evidence for the broader conclusions

R1.5a — three seeds and counts like 16/18

Most benchmark results use three seeds, while conclusions are drawn from counts such as 16/18 and 17/18. Report uncertainty where possible and use more cautious wording when the number of runs is small.

Response. Done, and the wording is now built around the intervals rather than the counts. A new Table 12 gives every pooled verdict count with a Clopper–Pearson 95% interval: $16/18 \to [0.65, 0.99]$, $18/18 \to [0.81, 1.00]$, $17/18 \to [0.73, 1.00]$. Mean columns carry across-seed standard deviations. Every claim in §5.8 is now stated as "16 of 18 runs, 95% CI $[0.65, 0.99]$" rather than as a bare fraction, and the model-axis discussion is rewritten (see R2.W6) to stop treating single-seed rows as evidence of stability.

R1.5b — SimNPO and another recent method

SimNPO is discussed in related work but not evaluated. Adding SimNPO and, if possible, another recent utility-preserving method would make the comparison more representative.

Response. SimNPO (Fan et al., 2025) is implemented and evaluated. We added it as a reference-free objective, $L = (2/\beta)\,\mathrm{softplus}(-\beta(\mathrm{nll}_\theta - \gamma))$ on the length-normalised forget NLL with $\beta = 0.1$, $\gamma = 0$ — NPO's objective with the frozen reference replaced by a constant margin — and ran it end to end on TOFU/GPT-2 under identical seeds, canary manifests and budgets as the existing subjects, so it is directly comparable. It certifies on all three seeds and achieves the widest forget/retain separation of any subject that remains fluent (forget NLL 6.20 against retain 2.62, versus NPO's 5.50 against 2.60), at a capability NLL of 4.17 against retraining's 3.23. It is reported in the new metrics table (Table 13) and named in §5.8 as the best-behaved of the approximate methods on the utility axis.

On a second method: RMU (Li et al., 2024) is now implemented and evaluated as well (new §5.9, Table 14). We chose it deliberately over task-vector negation because it is the one subject in the toolbox that never touches the token loss — it pushes the forget set's hidden activations toward a fixed random direction at one layer while holding the retain set's activations near the frozen reference's — so it tests whether the framework's conclusions survive a method of a different kind. Nothing in the audit changes to accommodate it, which is the point.

It is run co-trained with the other utility-preserving subjects and the two controls in a single run on TOFU/GPT-2, over the same three seeds as the main benchmark tier, so every row shares one fine-tuned adapter, one cohort and one machine per seed. RMU certifies at $\varepsilon = 0.2$ on all three seeds, with a realised cohort advantage of +0.050 — close to what retraining a fresh adapter achieves on the same cohorts (+0.026), against the un-unlearned model's +0.097 — at a retain NLL of 2.55 against retraining's 1.99.

The comparison also produced a finding we did not expect and now report, because it bears on your R1.5c point about what utility readings are for. RMU barely raises the forget split's likelihood at all (2.73 against the un-unlearned model's 2.19), whereas NPO and SimNPO raise it to 6.34 and 5.79 — yet all three land at a comparable realised advantage (+0.050, −0.014, −0.035). A forget-quality metric reading likelihoods would rank RMU far below the other two; the certificate treats them as comparable, because what it measures is whether an attacker can still *distinguish* a trained twin from its ghost. Which ranking is right depends on what the audit is for, and it is the clearest instance in the study of the two readings coming apart on real models. The un-unlearned control itself disagreed across seeds — revoked twice, issued once at a realised advantage of +0.042 — the same phenomenon §5.8 already reports for two cells of the main benchmark: $\varepsilon = 0.2$ is loose enough that a weakly-memorising run can certify without any unlearning having occurred.

The tier is also no longer single-architecture: the identical protocol repeated unchanged on TOFU/Pythia-160M (new §5.10, Table 15) reproduces the same pattern — RMU certifies at $\varepsilon = 0.2$ and 0.1 on all three seeds and turns undetermined at 0.05, at a realised advantage close to retraining's and a forget-NLL shift far smaller than NPO's or SimNPO's — so the loss/detectability divergence is not a property of one model family. Task-vector negation remains unevaluated, MUSE is untested for this method, and we still name the method list as incomplete.

R1.5c — held-out NLL is not sufficient for utility

Held-out NLL identifies severe collapse but does not establish general utility. Consider reporting standard TOFU/MUSE forgetting and retention measures together with at least one general capability benchmark.

Response. Agreed, and the added metrics changed what we can say. The end-to-end runner now records, behind an `--extra-metrics` flag and enabled for the revision-round runs: mean NLL on the benchmark's own *forget* and *retain* splits; greedy-decode ROUGE-L recall against the reference continuation on both splits, the TOFU/MUSE reporting convention; and a general-knowledge `capability_nll` on a fixed probe set — TOFU `world_facts`, or the MUSE holdout news stream. Results are in the new Table 13, on TOFU/GPT-2 over 3 seeds.

They sharpen the paper's central utility finding rather than softening it. Gradient ascent reaches forget NLL 94.3, retain NLL 93.7 and ROUGE-L recall of exactly 0.00 on *both* splits: it has not forgotten the forget set, it has stopped producing language. More usefully, the metrics reveal something held-out NLL alone had hidden: GradDiff's retain NLL is 6.8 but its capability NLL is 19.4, so its damage is nearly threefold larger *off* the retain distribution than on it. A retain-only utility reading — which is what we previously had — would have missed that entirely, which is a direct vindication of your point.

Two honest limitations. First, `world_facts` is a modest general-knowledge probe, not a suite such as MMLU; we say so in Appendix A and name a broader suite as the next step, preferring a small clearly described probe to implied coverage. Second, we discovered a confound while running this and report it rather than printing the flattering number: for TOFU the P1 relearn corpus *is* the capability probe set, so a post-P1 capability reading measures training on the probe. It reads 1.46, better than the retrained reference, which is an artefact. That cell is marked unavailable in Table 13 with the reason given.

R1.5d — systematic R-VOUCH

The R-VOUCH experiments would be more convincing if relearning, quantisation and jailbreak probes were evaluated more systematically across the main unlearning methods under clearly specified attack budgets.

Response. The budgets are now specified exactly, which was the more serious of the two gaps: Appendix A states P1 = 100 optimiser steps on a public corpus that never contains a canary, at batch 4 and learning rate 1×10^{-4}; P2 = symmetric per-channel round-to-nearest 4-bit quantisation of all linear and embedding weights; P3 = three fixed adversarial wrappers applied at scoring time only. These budgets are identical across every subject and model, which was true before but undocumented.

On systematic coverage across methods we have made partial progress and prefer to report it as such. The probes remain anchored on NPO in the benchmark tier, where P1 and P3 run on all six model/benchmark cells and all three seeds; P2 runs on the GPT-2 and TinyGPT tiers, and is skipped for the multi-billion-parameter models because the shared-frozen-base multi-adapter design would have to materialise merged weights to quantise them. Extending all three probes to all five unlearning methods multiplies the benchmark tier by roughly three in training cost, which was beyond this revision's budget. We have removed any implication that the probe study is exhaustive and state the coverage explicitly.

6. Related work

R1.6

The P1 relearning probe is closely related to Hu et al., ICLR 2025. The evaluation discussion would benefit from Wang et al., ICLR 2025 and Yuan et al., ICLR 2025, and Li et al., ICLR 2026 is particularly relevant to apparent forgetting and misleading automatic metrics. Please discuss these works in relation to VOUCH rather than only adding citations.

Response. All four are now discussed substantively, in a restructured §2 titled *Evaluating unlearning: a literature of negative results*, which we organised into three strands so that each work's bearing on VOUCH is explicit rather than implied.

Hu et al. (2025) is placed as the direct antecedent of our P1 probe: their demonstration that benign relearning on unrelated public text restores both hazardous knowledge and verbatim passages is precisely the obfuscation-versus-removal distinction P1 operationalises, and it is why we treat a post-relearning verdict as part of the certificate rather than an optional extra. We connect it to our own measurement: on the cohort target at $\varepsilon = 0.2$ the probe leaves NPO's certificate intact on every TOFU cell and on MUSE/Pythia, and *revokes* it on one MUSE/GPT-2 seed and one MUSE/Phi-1.5 seed — a change in 2 of 18 verdicts, both on the more open-domain corpus, where the public relearn set sits closest in distribution to the forget set.

Wang et al. (2025) contributes two findings we now act on: forget-quality metrics track surface behaviour and collapse under red-teaming, and the reported unlearning/retention trade-off is not a Pareto frontier, so method rankings are hyperparameter artefacts. We now state explicitly that this is why VOUCH does *not* rank unlearning methods against one another but certifies each against a declared tolerance with a paired utility reading.

Yuan et al. (2025) shows ROUGE-style matching hides degenerate models. We connect this to our own Figure 5, in which an NPO-unlearned model emits `LLLLLL...` on canary prompts while held-out loss is untouched — an instance of exactly their failure mode, caught here not by a fluency metric but by two-sided certification, which treats scoring conspicuously *below* the ghost twin as leakage.

Li et al. (2026) required the most careful treatment because it bears on the *limits* of what we certify, and we have written it that way rather than defensively. Their squeezing effect — probability mass pushed off the target and redistributed onto semantically equivalent rephrasings — means mass that has migrated to a paraphrase is, by construction, mass our default score class does not see, since our scores read the likelihood of the literal secret span. We state this plainly, note that the framework's response is structural (the class $\mathcal{F}$ is declared, the certificate is explicitly a statement about $\mathcal{F}$, and any named score can be added at the cost of power and no loss of validity since Theorem 2 needs no multiplicity correction), point to our out-of-class experiments (§5.13–§5.14) as a first step. We have also stopped calling a paraphrase-aware score the most valuable extension we have not made, and made it: §5.15 adds s_{para} — the *maximum* token-normalised log-likelihood of the secret over five paraphrase frames, rather than the mean over wrappers, since an average falls precisely when mass moves out of the frames it averages over — to $\mathcal{F}$ and reports the cost. It is one to two per cent in pairs and no verdict changes. We are equally explicit about what that does not establish: our secrets are ten random alphanumeric characters, and a high-entropy random string has no semantically equivalent rephrasing for mass to migrate onto, so this canary design cannot exhibit the squeezing effect. Testing that needs canaries whose *secret* is natural language, so that the twins are exchangeable as propositions rather than as strings, and that is the extension we now name.

We additionally added, on your and Reviewer 2's prompting, a new §3.2 *Measuring memorisation and forgetting* placing VOUCH in the secret-sharer lineage (Carlini et al., 2019) and the forgetting-measurement lineage (Jagielski et al., 2023). The comparison it draws is the useful one: exposure answers "how memorised is this string," whereas the ghost twin answers "could this have happened by chance," and the latter is what a certificate needs and what Zhang et al. (2025) show requires controlled randomisation.

7. Reproducibility and the cost comparison

R1.7a — what "retraining from scratch" means

Please define exactly what "retraining from scratch" means in the LoRA experiments. If this means training a fresh adapter from the same frozen base model without the forget data, the terminology should make this clear.

Response. Your reading is exactly right, and the terminology was misleading. Appendix A now opens with the definition: initialise a *fresh* adapter on the same frozen base and train it on D_{keep} only — neither organic forget data nor any canary — with the identical optimiser, schedule and step count as the original fine-tune. It is exact unlearning *within the adapter paradigm*: the adapter that saw the canaries is discarded, and the frozen base never saw a canary at any point. It is not retraining of pretraining, which we never perform. The subject is renamed **"retrain (fresh adapter)"** in every table.

We also answer, in the same appendix, R2.11's related question about whether adapter-level unlearning is representative of the deployment setting: the audit is indifferent, since it queries M_u as a black box, but a rank-16 adapter has fewer degrees of freedom in which to hide memorised content than a full fine-tune, so our subjects are certified in a regime that if anything favours them. We flag full-parameter runs as the natural next scaling step.

R1.7b — reproducibility settings

Please provide the key settings: exact model checkpoints, LoRA target modules, learning rates, training and unlearning steps, query counts, canary fractions, ordering procedure, tie-breaking rules, and stopping criteria.

Response. Appendix A now gives all of them: the eight exact model identifiers (including that Phi-1.5 is forced to fp32 because its LoRA fine-tuning diverges to NaN in fp16); LoRA rank 16, $\alpha_{\text{LoRA}} = 32$, dropout 0, `target_modules = all-linear`; the exact TOFU and MUSE splits and formatting; canary domains, repetition strata, secret entropy, per-tier cohort sizes and the measured canary character share; fine-tune/retrain at 600 steps, batch 8, AdamW 2×10^{-4}, block 160, clip 1.0, and every unlearning stage's steps and hyperparameters individually; $Q = 2$ wrappers for benchmark and model-axis runs and $Q = 4$ for the synthetic tier, with the score class per tier; the reveal order (a permutation seeded by the run seed, committed before any query); the tie-break PRNG seed (20260702, shared across all runs); the stopping criteria for both arms, including that the revocation arm deliberately continues after a crossing so the streaming product alarm stays well defined; and the seed counts for every simulation tier.

R1.7c — the cost comparison

Claims such as "roughly sixty times cheaper" or "two orders of magnitude cheaper" should be based on clearly matched hardware and workloads. Report the cost assumptions more explicitly.

Response. This criticism was well founded: the previous version quoted a single "seventy-six seconds" without naming the configuration and compared it to alternatives on unmatched hardware. Both are now fixed and the speed-up framing is abandoned.

Verification cost is stated as a workload: exactly $2Qm$ short forward passes and no training — 1,536 to 2,560 passes per subject at our settings. Measured wall-clock medians are reported *per tier, on the hardware that tier ran on*: 13 s (TinyGPT), 63 s (GPT-2, 640 pairs), 112–148 s (the 124M–1.4B benchmark tier), 397–499 s (the 3.8B–5.1B models). The comparison to descriptive alternatives is now a ratio of *workloads on the same model and hardware*: a retrain-based reference costs one full fine-tune, a shadow panel at least sixteen, against VOUCH's zero; on our TOFU/GPT-2 configuration one fine-tune is 600 steps at batch 8, about 4,800 sequence forward-and-backward passes, so a sixteen-model panel is roughly 77,000 training passes against VOUCH's 1,536 inference passes. Appendix A states the heterogeneous hardware explicitly and explains why we give a ratio rather than a speed-up factor: the two workloads are not interchangeable.

Minor comments

R1–M1 — GDPR and EU AI Act

Some statements about the GDPR and EU AI Act are broad. Please cite appropriate primary sources or use more careful wording.

Response. Rewritten against the primary instruments, both now cited with their Official Journal references and CELEX identifiers (European Parliament and Council of the European Union, 2016, 2024). The wording is corrected in two ways that matter. First, Article 17 GDPR creates the erasure *right* but contains no demonstrability requirement; the duty to be able to *show* compliance lives in Articles 5(2) and 24(1), with supervisory audit powers in Article 58(1)(b), and we now attribute it there. Second, we no longer write that the AI Act has "audit provisions," which overstates it: the Act imposes automatic event logging (Article 12), technical documentation and ten-year retention (Articles 11, 18), production to authorities on reasoned request (Article 21), conformity assessment (Article 43), and gives the AI Office evaluation powers over general-purpose models (Article 92). The paragraph now closes by noting that neither instrument prescribes a statistical test, which is the actual gap the paper addresses.

R1–M2 — promotional phrasing

Phrases such as "dissolves all three obstacles," "forgetting by lobotomy," and "what a regulator cares about" are somewhat promotional for a journal paper.

Response. All three removed, along with others we found on a sweep: "dissolves" → "addresses"; "forgetting-by-lobotomy" → "destructive unlearning"; "it is verification, not optimisation, that a regulator cares about" → "Evaluation, not optimisation, is the bottleneck," now supported by citations rather than assertion. We also toned down "impossible to miss," "emphatically," "straw man" and "embarrassment" in the experiments section.

R1–M3 — dense figures and tables; Table 7 seed counts

Some figures and tables are dense and use small text. Table 7 should make it immediately clear which models use one seed and which use three.

Response. The model-axis table (Table 7 in the previous version, now Table 18) carries an explicit **seeds** column, and its surrounding text is rewritten to distinguish the three-seed rows from the single-seed ones instead of describing the whole axis as stable (see R2.W6). We split the former single benchmark table into three narrower ones — verdicts, the tolerance sweep, and the interval-estimated counts — which removes the worst of the density, and the new tables are set at `\small` rather than `\tiny`. The benchmark verdict table remains the densest object in the paper at `\tiny`; it carries six model/benchmark cells and we could not find a narrower layout that keeps them side by side, which is the comparison the table exists to support.

Reviewer 2

We are grateful for a report that engaged with the construction closely enough to find the Table 3 censoring artefact and to press on the $\varepsilon = 0.2$ default. Both were right.

W1. Every end-to-end result uses $\varepsilon = 0.2$

R2.W1 / Point 1

$\varepsilon = 0.2$ means an adversary can still distinguish the in-twin from its ghost about 60% of the time. Tighter tolerances are never run on an actual language model. Report end-to-end results on at least one real model at $\varepsilon \leq 0.05$, or state explicitly in the abstract, introduction and conclusion that all end-to-end certification is at $\varepsilon = 0.2$, and discuss what that tolerance means operationally. Guidance on how to choose ε would also help.

Response. We have done the first of the two, and also the discussion you asked for.

End-to-end at $\varepsilon \leq 0.05$ (Table 11). The corrected finite-cohort target (R1.1) is materially more powerful than the fixed-boundary one, and replaying every saved run at $\varepsilon \in \{0.05, 0.1, 0.2\}$ now yields real-model certificates at $\varepsilon = 0.05$: exact unlearning is issued on all three seeds at $\varepsilon = 0.05$ for TOFU/Pythia-160M and TOFU/Phi-1.5, and on 12 of 18 benchmark runs overall. At $\varepsilon = 0.1$ it is issued on all 18.

And on the super-population target. The result above is a *cohort* certificate, and at 384–640 pairs the super-population arm certifies nothing below $\varepsilon = 0.2$ — Theorem 4 says it should not, since $\varepsilon = 0.1$ needs about 600 pairs. So we ran the experiment your point implies: a new tight-tolerance cohort of 3,072 pairs on TOFU/GPT-2, two seeds, with every canary inserted *once* ($r = 1$) to hold the corpus share to 22.5%. On that cohort the super-population target certifies at $\varepsilon = 0.1$ for all four subjects and both seeds — your request met on the target you asked about. At $\varepsilon = 0.05$ it is still undetermined, and we now say why: the 2,400-pair figure is for a one-sided crossing, and two-sided certification costs about 1.7×, so $\varepsilon = 0.05$ wants roughly 4,000 pairs. Both targets are reported side by side in Tables 11 and 16 and we recommend reading them together.

That experiment also produced the most useful negative result in the revision, and we would rather lead with it than bury it. At $r = 1$ this model memorises essentially nothing: realised advantages span -0.040 to $+0.025$, and the mean score gap for the *un-unlearned* model is $+0.005$ nats. So "no unlearning" certifies at $\varepsilon = 0.05$ alongside retraining — correctly, since on that cohort there is no residual advantage to find. A single exposure of a single canary in a 6,872-document corpus leaves no measurable trace in a 124M-parameter model. This is the sharpest form of your W2 point that we can state: the audit's power at the most organic-like dose is not merely lowest, on this model it is nil, so a certificate issued there is uninformative rather than wrong. We have added a subsection saying so (§5.11) and named it in the conclusion as the most important open problem the framework faces.

What $\varepsilon = 0.2$ means, said plainly. §5.8 now closes with an explicit paragraph on choosing ε, which states that $\varepsilon = 0.2$ licenses a 60% distinguishing rate and $\varepsilon = 0.05$ a 52.5% one, and gives three considerations that set it: a regulatory one (if the audit must support a demonstrability claim, the permitted advantage should not itself be reportable as a privacy loss, arguing for $\varepsilon \leq 0.05$); a statistical one (Theorem 4 fixes the cohort at $\approx 2\log(1/\alpha)/\varepsilon^2$ pairs, so ε is bought with canaries — 600 at 0.1, 2,400 at 0.05, 15,000 at 0.02); and an operational one (the cohort must stay a negligible corpus share, which makes $\varepsilon = 0.02$ affordable for a provider with millions of records and unaffordable at benchmark scale). We now describe our own $\varepsilon = 0.2$ as a *demonstration* tolerance chosen so that 384–640 pairs suffice on academic corpora, explicitly not a recommendation, wherever the number appears.

One further consequence we surface rather than hide. Under the corrected, more powerful target, the two GPT-2 cells in which the un-unlearned model was previously "undetermined" are now *certified* at $\varepsilon = 0.2$ — their realised advantages are 0.062 and 0.078, genuinely below the tolerance. So at $\varepsilon = 0.2$ a model that was never unlearned at all can pass. That is the sharpest available illustration of your point about the permissiveness of the default, and we now use it as such in §5.8.

W2. Canaries versus a user's deleted record

R2.W2 / Point 2

The certificate concerns canaries; the motivating scenario concerns a user's deleted record. The introduction opens with GDPR Article 17 and the abstract makes no qualification. The r strata calibrate but do not license the extrapolation, and the $r = 1$ end is where the measured gap is smallest. Either give a transfer statement under an explicit named assumption, or reposition the contribution as a certificate about a planted cohort. Also clarify that the guarantee is per-cohort, not per-request.

Response. We have repositioned, and additionally supplied a transfer statement — but for the other transfer, since the two should not be conflated.

Repositioning. The paper now says throughout that the certificate is about the committed canary cohort. The abstract's closing sentence names all three qualifications. The introduction's scope paragraph states that the guarantee is per-cohort, *not per-deletion-request*, because per-request certificates would need per-user canaries — and this now appears in the introduction, not only in the scope paragraph of §4 and the conclusion. The legal framing is rewritten (R1–M1) so that it no longer implies the certificate discharges an Article 17 obligation for a particular data subject; it presents the accountability duty as the reason evidence is needed at all.

The transfer statement we can prove. Proposition 3 bridges the *realised cohort* advantage to the *super-population* advantage over the coin draw, under an explicitly named bounded-coin-influence assumption, at cost $\kappa\sqrt{2\log(2/\alpha')/m}$. That is a genuine, named, quantified transfer.

The transfer we cannot prove, stated as such. The step from planted canaries to *organic* records is a different matter and we do not have a theorem for it. We say so, and we have strengthened the empirical calibration and its stated limits: the dose–response curve is now reported with dispersion, and we state explicitly that $r = 1$, the stratum most like an organic record, carries the weakest signal, so the extrapolation is weakest exactly where it matters most. §5.12's head-to-head against forget-quality and min-k% on identical runs (see R1.2) is the most direct evidence we can offer that canary verdicts track conventional forgetting measurements, and we present it with its limitations rather than as a substitute for the missing theorem.

W3. Proposition 6 runs one way only

R2.W3 / Point 3

Proposition 6 shows $(\varepsilon_u, \delta_u)$-certified algorithms pass VOUCH. There is no converse, so passing VOUCH implies no $(\varepsilon_u, \delta_u)$ guarantee. Calling it a "bridge" and "semantic anchor" invites the wrong reading. State the one-directionality immediately after the proposition.

Response. Done, as Remark 4 placed immediately after the proposition, and stated at least as strongly as you put it: "The converse is false and we claim nothing like it: passing VOUCH does *not* imply any $(\varepsilon_u, \delta_u)$ guarantee, does not bound the influence of any datum outside the canary cohort, and does not constrain scores outside $\mathcal{F}$." The remark also reframes the proposition's purpose as a consistency check on the definition — an algorithm with a parameter-space guarantee should not be flagged by a behavioural audit — and says it should not be read as a semantic equivalence in either direction. The phrase "semantic anchor" is deleted; §2 now calls it a consistency check and cross-references the remark. The proposition is also retitled "One-directional bridge from certified unlearning" so the qualification travels with the name.

W4. The declared class $\mathcal{F}$ is never stress-tested from outside

R2.W4 / Point 4

The certificate is a supremum over $\mathcal{F}$, so a stronger attack outside $\mathcal{F}$ could exceed ε on a certified model. The R-VOUCH probes re-run the same score class, so they do not test this. Run at least one attack outside $\mathcal{F}$ against a certified model and report whether the residual advantage *stays below ε. If infeasible, say why, and soften the claim that VOUCH bounds "the largest residual advantage any score in the class can extract."*

Response. We ran the experiment, it produced a partial negative result, and we report it as such (§5.13, Table 18). Two attacks that no member of $\mathcal{F}$ implements were run against all 178 model/subject/seed combinations certified at $\varepsilon = 0.2$:

- *stratum-restricted*: the repetition label r_i is part of the committed manifest, so a verifier may condition on it; restricting the audit to $r = 8$ concentrates it on the most-memorised pairs. No score in $\mathcal{F}$ uses r_i.
- *cross-fitted learned combination*: a Fisher direction over the declared scores estimated on the first half of the reveal order and applied to the second. A trained combination of members of $\mathcal{F}$ is not itself a member of $\mathcal{F}$.

Both preserve Theorem 1 — the first conditions only on committed per-pair data, the second uses a predictable weight vector — so both come with anytime-valid bounds rather than point estimates.

attack	runs	pairs	observed $> \varepsilon$	binomial null	mean CS U_t
stratum-restricted ($r = 8$)	178	96–159	0.028	0.035	0.078
cross-fitted learned combination	184	192–1536	0.005	0.003	0.063

We must report this as a *null* result, and we say so because our own first reading of it was wrong. The raw exceedance rates (2.8% and 0.5%) look like breaches until one asks what a *clean* subsample of the same size would produce: these attacks are evaluated on 96–159 and 192–1,536 pairs, and an exact binomial calculation puts the chance exceedance rate at 3.5% and 0.3%. The stratum-restricted attack therefore exceeds the tolerance *less* often than noise predicts, and the learned combination's single exceedance in 184 runs is within a coin-flip of its own null. The anytime confidence bounds agree: their means, 0.078 and 0.063, sit far below ε.

So the experiment answers your question in the negative but does not settle it. It does not show that $\mathcal{F}$ is too narrow — the two attacks we could build produce no signal beyond subsampling noise — and it does not license the converse, because both attacks are weak: one is a subpopulation restriction that pays for its focus in sample size, the other a linear recombination of scores already in $\mathcal{F}$. Our position is now that the certificate's dependence on $\mathcal{F}$ is a *definitional* limit, visible in the theorem's statement rather than in our measurements, and that we have not found an attack strong enough to put a number on it.

We have accordingly softened the claim exactly as you ask. The phrase "the largest residual advantage any score in the class can extract" is gone; the guarantee is written as a statement about $\mathcal{F}$ in the abstract, the introduction's scope paragraph, the scores subsection, the scope paragraph of the theory section and the conclusion. We are also careful not to over-read the table in the other direction: a null result does not establish that the class restriction is immaterial, which is why §5.14 goes on to run an attack strong enough to say something.

What the design does offer, and what we now say instead, is that enlarging $\mathcal{F}$ costs power and no validity — any score a sceptic names can be added, and the certificate theorem needs no multiplicity correction — so the framework absorbs a stronger attack with no change to the theory. We flag two directions. The subpopulation reading is worth keeping independently of this table: the certificate bounds a cohort *average*, and our dose–response curve reports realised advantages of $+0.73$ at $r = 8$ against $+0.22$ at $r = 1$, so an average below the tolerance is compatible with a stratum above it; certifying per stratum closes that at the cost of multiplying the cohort requirement by the number of strata. And the most promising single addition is a paraphrase-aware score, since Li et al. (2026) show that gradient-ascent objectives move mass onto semantically equivalent rephrasings — mass our literal-span scores cannot see by construction.

And, since that draft, the LiRA-style attack you suggested. You were right that it was the experiment worth running, and we have now run it (new §5.14, Table 19). It needs the whole pipeline repeated once per shadow, which the 124M–5.1B tiers cannot afford here, so it runs on the TinyGPT tier — the one the paper already uses where retraining is cheap enough to be ground truth — with 16 matched shadows and a cohort of 256 pairs. Each shadow repeats the pipeline with an independent redraw of the inclusion coins, so every twin is trained on in about half of them, and the unlearned subject is calibrated against shadows that were themselves unlearned.

The positive control is what makes the result readable, and we would draw your attention to it first: on the un-unlearned model LiRA achieves $\hat{\Delta} = +0.820$ (95% CI $[+0.736, +0.884]$) and agrees with the declared class's own guess on 92% of pairs. The attack works. On the NPO model VOUCH certifies at $\varepsilon = 0.2$, the same attack yields $\hat{\Delta} = -0.016$, 95% CI $[-0.141, +0.110]$, with agreement falling to 0.47 — chance. The interval excludes every tolerance we certify at.

This is a different kind of evidence from the two attacks above, and we are careful about how much it carries. It is one target model, one tier, one unlearning method and 16 shadows against the hundreds a canonical LiRA would use, so it does not show that $\mathcal{F}$ is wide enough in general. But unlike the stratum-restricted and learned-combination attacks, it cannot be dismissed as too weak to have found anything: it finds a great deal where leakage exists and nothing where the certificate says there is none. That is the first measurement in the paper that *bounds* the class-restriction gap rather than merely naming it.

W5. No comparison against an existing verification method

R2.W5 / Point 5

Section 5.5 asserts that VOUCH's alarm catches a GradDiff failure "that a forget-metric evaluation would have passed," but that comparison is never run. Adding a column showing what forget quality and min-k% conclude on the same models and seeds would convert the paper's central rhetorical claim into evidence, and it is cheap to do.

Response. Agreed, and it was as cheap as you expected. §5.12 and Table 17 now report both metrics on the same models and the same seeds, and the specific claim you flagged is now substantiated: on the MUSE/Phi-1.5 GradDiff seed that VOUCH *revokes*, the min-k% leakage analogue reads -2.4% (-1.2% averaged over that cell's 3 seeds) — a *negative* leakage figure on a model carrying real residual memorisation. That is the evidence the previous version asserted.

Two further disagreements appear, and we report them even though one is unflattering to the descriptive metrics and the other is unflattering to nobody in particular. The forget-quality analogue is systematically pessimistic on this data, failing genuine unlearning methods whose realised advantage is within noise of zero, because a KS test on several hundred paired scores detects distributional differences far smaller than any tolerance one would certify. And the min-k% statistic inspected after every pair — the way an auditor facing a stream of deletions would naturally use it — flags leakage on genuinely clean retrained models in several cells, which is optional stopping doing what §5.2 predicts.

We are also explicit about the comparison's limits: these are analogues computed on canary cohorts, not the benchmarks' own metrics on their own splits, and a metric designed for one target should not be judged too harshly on another. The narrow point stands: on identical data the descriptive and certified readings disagree in both directions.

W6. Thin statistical reporting

R2.W6 / Point 6

Three seeds per cell, bare I/R/U patterns, no interval estimates, no dispersion on the mean columns. Counts such as "sixteen of eighteen" carry wide binomial uncertainty at $n = 18$. Table 7 appears to use a single seed for the larger models and shows a markedly weaker picture, yet the text describes the axis as "stable". The honest reading is that the framework returns undetermined at the cohort sizes used for the models that matter most.

Response. Every part of this is addressed, and we accept the characterisation in the last sentence.

Intervals and dispersion. New Table 12 gives every pooled verdict count with a Clopper–Pearson 95% interval (16/18 $\to$ $[0.65, 0.99]$; 18/18 $\to$ $[0.81, 1.00]$; 17/18 $\to$ $[0.73, 1.00]$). The

GPT-2 end-to-end table gains an across-seed standard deviation column on $\bar{D}$. Counts in the prose are now always given with their interval.

Table 7 and the "stable" claim. The table now carries an explicit **seeds** column making the single-seed rows unmistakable, and the surrounding text is rewritten. It no longer says the axis is stable. It says: detection is stable — every un-unlearned model is revoked at every scale, on every seed run; certification is *not* uniformly stable — at the 384-pair cohorts used for the largest models the super-population target returns undetermined for retraining and NPO on Qwen3-0.6B and Qwen3-4B, exactly as Theorem 4 predicts at that cohort size; and one seed per cell is too thin to distinguish a systematic effect from noise. We also state that those cells do issue under the cohort target, so the reader can see which claim is cohort-size-limited and which is not.

W7. Table 3's starred entries are selection-biased

R2.W7 / Point 7

Starred medians are taken over runs that issued within a 20,000-pair horizon, so both starred entries sit below the information-theoretic bound, which cannot be right and is an artefact of censoring. As printed, the row reads as though VOUCH beats the KL limit. Report the censoring rate and either use a survival estimate or mark them as lower bounds.

Response. This was a genuine error and your diagnosis of it is exactly correct; an entry below the information limit should have been a red flag to us. The power table now reports the censoring rate and a Kaplan–Meier median, and the artefact disappears:

ε	median | issued	Kaplan–Meier median	censored	$\log(1/\alpha)/\mathrm{KL}$
0.02	11,189	18,377	**0.45**	14,976
0.05	2,998	2,998	0.00	2,394
0.10	686	685	0.00	596
0.20	166	165	0.00	147

At $\varepsilon = 0.02$, 45% of runs are censored and the corrected median is 18,377, comfortably *above* the limit of 14,976; conditioning on the fast half had produced 11,189. At $\alpha = 0.01$ censoring reaches 72% and the survival curve never reaches one half, so no median exists and we now print "—" rather than a number. The caption states the censoring convention, and §5.5 explains the artefact explicitly so a reader of the earlier version can see what changed.

One presentational note. The table also has a finite-cohort column whose entries can legitimately sit below the KL bound, because the cohort claim resolves deterministically once the cohort is exhausted and so is capped by m rather than governed by a super-population information limit. We flag this in both the caption and the text so that the column is not misread the way the starred entries were.

Point 8. Are the canaries detectable?

R2.8

The canaries are templated with fictitious entities and 40 bits of entropy. The paper asserts they "do not stand out" but never tests this. Please report a detection experiment — for example, whether a perplexity filter or a small classifier can separate canaries from organic corpus records — since a provider who can identify the committed cohort defeats the audit, and the honest-but-verifiable threat model is doing a lot of work here.

Response. We ran the experiment and it contradicts what we had written. We retract the claim.

A perplexity filter under the *pristine base model* — one forward pass per record, no knowledge of the template — separates the deployed canaries from the TOFU corpus with AUC 1.000 and recovers the *entire* cohort at a 1% false-positive rate (§5.16, Table 21). A trained character-n-gram classifier does the same.

There is an information-theoretic reason, and stating it turns the negative result into a design rule. A span carrying H bits of entropy that the scoring model has never seen costs at least $H \ln 2$

nats of total surprise, so a secret packed into T tokens must raise the record's mean token NLL by at least $H \ln 2/T$ above the organic baseline. Our deployed template spends 51.7 bits over 25 tokens: a predicted excess of 1.45 nats per token against an organic standard deviation of 0.42. Detection was not a failure of craft but an arithmetic certainty.

template	bits	tokens	predicted excess	ppl. AUC	recall @ 1% FPR
deployed (alphanumeric)	52	25	1.45	1.000	1.00
word-composed	42	35	0.84	1.000	1.00
entropy-diluted	68	88	0.54	0.974	**0.21**

The rule identifies the lever as T, not the vocabulary — re-spending the budget on natural words changes nothing — and acting on it works: an entropy-diluted template carrying *more* entropy (68 bits) over 88 tokens cuts the filter's recall from 1.00 to 0.21. Both templates are in the released generator. We keep two caveats visible: a *trained* detector still recovers every cohort, because the template's fixed phrasing is a signature — though a provider who can fit one already knows the template, a far stronger position than the perplexity filter needs and one the threat model already excludes; and dilution presumably costs power, since a longer span dilutes the memorisation signal along with the entropy, which we have not yet measured end to end and therefore do not present as a drop-in replacement.

On your framing that a provider who identifies the cohort "defeats the audit", we would refine it slightly, because the distinction turns out to be the crux of the threat model. Both twins in a pair are drawn from the same template distribution and are therefore *equally* conspicuous, so a provider who locates the cohort learns which records are canaries but nothing whatever about which twin the coin admitted: the null of Theorem 1 survives complete disclosure of the manifest's *contents*. What it does not survive is a provider who, having located the cohort, treats it differently from organic data. That is special-casing; it is what the honest-but-verifiable assumption excludes; and it is precisely the gap cryptographic attestation exists to close. We now state that assumption wherever the certification claim is summarised, and say in the conclusion that we now know it is easier to violate than we had supposed.

Point 9. Query count Q and the cost figure

R2.9

Report the number of query prompts Q and how they were chosen. Q affects both power and the 76-second cost figure and is currently unspecified. State which configuration produced the 76 s number, and give cost as a function of ε and model size.

Response. Q is now stated wherever it matters. $Q = 2$ for every benchmark and model-axis run (the identity wrapper and "According to the archives, ...") and $Q = 4$ for the GPT-2 synthetic tier, which also carries the base-calibrated `ratio` score; Appendix A lists the wrappers and the per-tier score class. The wrappers were chosen as light paraphrases of the canary prefix that preserve the secret span's position, and Q was set to 2 in the benchmark tier for cost reasons, which we now state rather than leave implicit.

On the cost figure: the previous "seventy-six seconds" is withdrawn rather than re-attributed, because it did not correspond to a configuration we could identify unambiguously in the logs, and quoting it would compound the original error. Cost is now given as a closed form plus per-tier measurements. The workload is exactly $2Qm$ forward passes and no training. As a function of the two variables you name: m scales as $2\log(1/\alpha)/\varepsilon^2$ by Theorem 4, so verifier cost scales as $4Q\log(1/\alpha)/\varepsilon^2$ forward passes, and the per-pass cost scales with model size. Measured medians, per tier and on the hardware that tier ran on, are 13 s (TinyGPT, 0.9M), 63 s (GPT-2, 640 pairs), 112–148 s (124M–1.4B benchmark tier) and 397–499 s (3.8B–5.1B). Appendix A records the heterogeneous hardware and explains why we report workload ratios rather than speed-up factors.

Point 10. The tie rate

R2.10

State the tie rate for $D_i = 0$ empirically. With 4-bit quantised models and degenerate NPO outputs, near-ties may be systematic rather than negligible, and the committed-coin fix deserves a measured justification.

Response. Measured across every scored pair in the study, and the answer is more interesting than we expected, in the direction you anticipated. Across all 124,416 scored pairs and for s_{loss}, exact ties occur in *zero* pairs, at every architecture from 0.9M to 5.1B parameters, including the degenerate-output NPO runs. For s_{mink} there is one exception and it is exactly where you predicted one: 2.7% of pairs (42 of 1,536) tie exactly on Gemma-4-E2B, because the min-k% statistic over a short span takes few distinct values under that model's quantised inference path.

So the committed tie-break coin is a formality for the loss score and is load-bearing for min-k% on that model — which is the measured justification you asked for, and we report both numbers in §3.4 and §5.8 rather than only the reassuring one. We also report the near-tie rate at $|D| < 10^{-6}$ in the released re-analysis JSON.

Point 11. LoRA adapters and the deployment setting

R2.11

Clarify that all fine-tuning and unlearning is applied to LoRA adapters of rank 16, and discuss whether adapter-level unlearning is representative of the deployment setting the certificate is meant to cover.

Response. Stated in the experiments preamble, in Appendix A, and in the definition of "retrain (fresh adapter)" (R1.7a). On representativeness we give a direct two-part answer rather than a disclaimer. The *audit* is indifferent: it queries M_u as a black box and never inspects parameters, so nothing in Theorem 1 or Theorem 2 changes if the provider unlearns by full-parameter gradient steps. What the adapter setting changes is the *difficulty of the problem the subjects face*: a rank-16 adapter has far fewer degrees of freedom in which to hide memorised content than a full fine-tune, so our subjects are certified in a regime that if anything favours them, and a provider doing full-parameter unlearning should expect verdicts at least as demanding. We flag this as a limitation of the empirical study rather than of the framework, and name full-parameter runs as the natural next scaling step.

Point 12. Framing the 99.6% figure

R2.12

The 99.6% figure is a comparison against the authors' own earlier design under an adversarially constructed composite null, not a refutation of any published method. The abstract currently reads as the latter.

Response. Correct, and corrected. The abstract now reads "the adaptive-mixture alternative of our own earlier design is anytime-invalid, falsely certifying 99.6% of the time under an adversarially constructed composite null." Both attributions — ours, and adversarial — are attached to the figure at every one of its four occurrences: the abstract, the introduction's list of corrections, the remark on why the mixture fails ("which is our own earlier design, not a published method"), the experiments section, and the caption of the soundness table.

Minor comments

R2–m1 — secret sharer and forgetting measurement

The canary methodology descends from the secret-sharer line, and the forgetting-measurement literature is closely related; neither is cited. Please add Carlini et al. (2019) and Jagielski et al. (2023) and situate the ghost-twin design relative to them.

Response. Both added, with a new §3.2 devoted to situating the design. Carlini et al. (2019) is credited with the planted-canary device and the exposure metric, and we state the one change that matters for inference: exposure answers "how memorised is this string," whereas randomising which twin enters training makes the measurement carry a *null*, and the ghost twin answers "could this have happened by chance" — the question a certificate must answer and the one Zhang et al. (2025) show requires controlled randomisation. Jagielski et al. (2023) is credited with measuring forgetting via the decay of attack success, and we note that their two findings — that models do forget, but that non-convex models can retain indefinitely and that deterministic training exhibits no forgetting at all — are the premise of ours: incidental forgetting is real but uncontrolled, and so cannot substitute for a certificate. Bărbulescu and Triantafillou (2024) is connected to the same discussion as the per-sequence memorisation-strength result our repetition strata traverse.

R2–m2 — numerals

Use numerals rather than words for quantities throughout ("sixteen of eighteen", "seventy-six seconds", "six hundred").

Response. Done throughout, by sweep and manual check: "16 of 18", "600 pairs", "2,400", "2,000 seeds", "+0.46, +0.78, +1.54, +2.35 nats at $r = 1, 2, 4, 8$", and so on. The seventy-six-second figure is withdrawn entirely (R2.9). Number words are retained only where they are ordinary English rather than reported quantities ("one of the two options", "three obstacles").

R2–m3 — narrative register and Section 1 length

The register in Sections 1 and 5 is engaging, but Section 1 buries the contribution list under three pages of motivation. Consider tightening.

Response. The introduction is restructured with signposted paragraphs — *Three obstacles*, *The design*, *What the certificate claims, and what it does not*, *Corrections carried by this version*, *Contributions* — so a reader reaches the claims quickly and can navigate to the scope statement directly. We should be candid that the introduction is not shorter: it has grown from about 860 to about 1,380 words, because it now carries the scope statement both reviewers asked for (*What the certificate claims, and what it does not*), the list of five corrections, and the sourced legal framing of R1–M1. What has been cut is expository motivation: the three-obstacle discussion is tightened and the evaluation-critique literature now carries the motivational weight in two sentences with citations rather than in expository prose, so the *Contributions* paragraph closes a two-page introduction rather than following three pages of motivation. We have kept the narrative register in the experiments, including the development anecdote about the NPO sign error, because it carries a point about what statistical certification buys that we could not make as economically any other way — but we have removed the more self-conscious framings ("One episode during development is worth recounting" is now a plain topic sentence).

R2–m4 — conservatism is not tightness

A realised error of 0.031 against a nominal 0.05 is expected of a supermartingale and is not itself evidence of tightness; the conservatism also costs power. Worth a sentence acknowledging this.

Response. Added, in §5.2 immediately after the calibration numbers: "A realised error of 0.031 against a nominal 0.05 is what a supermartingale bound is *supposed* to produce; it is evidence of validity, not of tightness, and the same conservatism costs power, which is the price Table 5 quantifies. We do not claim the bound is tight and the gap is not a defect." The cross-reference is to the new sequential-baselines table, so the reader can see the cost measured rather than merely conceded.

R2–m5 — Fig. 2's left panel duplicates the text

Fig. 2's left panel and the corresponding text make the same point twice; one could go.

Response. The soundness result is now carried by the table and Remark 2 alone, and the redundant figure panel is dropped from the manuscript.

R2–m6 — Theorem 3's statement

Theorem 3's statement should specify that $\mathbb{E}[\tau^*]$ *is under exact unlearning and small* ε*, as the proof's final step assumes.*

Response. Done: Theorem 4 now reads "…the expectation being taken under exact unlearning and the asymptotics in small ε", and the statement additionally notes that for the finite-cohort target the time is capped by the cohort size m, since the claim resolves deterministically once the cohort is exhausted.

R2–m7 — typo in Section 5.3

"$\Delta = 0.1$ *in six hundred and ninety" — please check against Fig. 3.*

Response. Checked against the released simulation output: the value is 690 pairs, which matches Figure 3. The sentence now reads "$\Delta = 0.1$ in 690", in numerals.

A note on what this revision does not settle

Since the first revision draft we have closed five of the six gaps this section previously listed: the LiRA-style attack is run (§5.14), RMU is added as a second utility-preserving subject (§5.9) and its tier brought to the same three seeds as the main benchmark tier, the paraphrase-aware score is implemented and added to $\mathcal{F}$ (§5.15), Theorem 5 supplies the finite-population power analysis, and the reveal order can now be drawn from a public randomness beacon. What remains is narrower than before, and sharper for it.

1. **The LiRA result's reach.** It is one target model, on the TinyGPT tier, against one unlearning method, with 16 shadows. Its positive control establishes that the attack works, so its null on a certified model is informative rather than vacuous — but a shadow study at 124M parameters and above needs the whole pipeline repeated per shadow, which we could not afford, and nothing here rules out a larger model behaving differently.
2. **The squeezing effect is still untested.** We added the score Li et al. (2026) motivate, and it costs almost nothing to carry, but our secrets are random alphanumeric strings and a random string has no paraphrase. Testing whether mass migrates onto rephrasings needs canaries whose secret is natural language — twins exchangeable as propositions, not as strings — which is a canary design we have not built.
3. **Method and probe coverage.** SimNPO and RMU are added; task-vector negation is not. The R-VOUCH probes remain anchored on NPO in the benchmark tier, P2 runs on the small tiers only, and the capability probe is `world_facts` and the MUSE holdout rather than a suite such as MMLU.
4. **Theorem 5 bounds one of the two mechanisms.** It gives an exact, strategy-independent finite-sample guarantee via the null's remaining feasible compositions, and that is what was missing. It does not give a growth rate for the wealth process under without-replacement reveal, which is usually the mechanism that fires first; the cohort target's typical timing is still reported empirically.
5. **The released numbers predate the beacon.** Every table in the paper was computed with `reveal_source = "seeded"`, in which the reveal order is derived from the run seed and so does not meet Theorem 2's independence hypothesis. The two sources that do meet it are implemented and tested, and a deployed audit should use the beacon; we report the tables as they were computed rather than restating them under a reveal rule they were not run under.
6. **The RMU tier is TOFU only.** Three seeds, matching the main benchmark tier, but reported separately rather than merged into it, and now checked on two architectures (GPT-2 and Pythia-160M, §5.9–§5.10) with the same pattern on both; MUSE and larger models remain untested for this method.

Set against those, the changes we regard as materially strengthening the paper are the ones the reviewers forced: a certificate whose validity no longer depends on an independence assumption that our own data violates, a tolerance sweep that reaches $\varepsilon = 0.05$ on real models with the cohort/super-population distinction made explicit, a comparison against valid sequential procedures rather than a straw baseline, a head-to-head against the metrics we claim to replace, a detectability measurement that overturns a claim we should never have made without testing, and — added since the first draft — an attack from outside $\mathcal{F}$ that is strong enough for its null to mean something. We are grateful for both reports.

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